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Exoskeletons: History, Commercialisation, Industry Landscape and Future Outlook

K Vijay Anand 25 October 2026

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Abstract

Powered and passive exoskeletons have moved from a 1890 patent and a failed 1960s military prototype to a commercial industry with FDA-cleared medical devices, tens of thousands of industrial units in service, and fielded military load-carriage trials. This paper traces that history in four eras, explains the enabling technologies, and examines how the first products crossed the regulatory, reimbursement and standards barriers to market. It then maps the major manufacturers in 2026 across medical rehabilitation, personal mobility, industrial, military and consumer applications, with product links. Finally it reviews market estimates and the open technical problems of power density, human–machine interaction, fit and safety, and outlines the likely direction of each application segment over the next decade, including opportunities for Indian defence and industry.

Keywords: exoskeleton, exosuit, wearable robotics, rehabilitation robotics, human performance augmentation, load carriage, ISO 13482, ASTM F48

1. Introduction

An exoskeleton is a wearable mechanical structure that acts in parallel with the human body to support, assist or augment movement. Unlike a prosthesis, which replaces a missing limb, or a stand-alone robot, which works apart from the person, an exoskeleton shares load and motion with the wearer's own joints. Its output therefore depends on how well it reads and follows human intent.

Three needs drive the field:

- **Restoring mobility** to people with spinal cord injury (SCI), stroke, multiple sclerosis or age-related weakness, and giving therapists a way to deliver high-repetition gait training.
- **Preventing injury** in workers who lift, bend or hold tools overhead for long shifts. Work-related musculoskeletal disorders remain among the largest causes of lost workdays in manufacturing, logistics and construction.
- **Extending human endurance and strength** for soldiers, first responders and anyone who must carry heavy loads over distance.

This paper covers the classification of exoskeletons (Section 2), their history (Section 3), enabling technologies (Section 4), the path to commercialisation (Section 5), the major players in each application area as of October 2026 (Section 6), the market (Section 7), open challenges (Section 8) and the future outlook (Section 9).

2. Classification of exoskeletons

Exoskeletons are classified along four independent axes. Any one device sits at one point on each.

Axis	Categories	Typical examples
Body region	Upper limb / shoulder; lower back (lumbar); lower limb (hip, knee, ankle); full body; hand / glove	Shoulder tool-support vests; lumbar lift-assist suits; gait-training legs; full-body heavy-lift suits
Actuation	**Passive** (springs, elastic bands, gas struts store and return energy); **Active / powered** (electric motors, hydraulics, pneumatics, with battery and controller); **Hybrid / quasi-passive** (passive elements with small actuators or clutches that lock, release or tune them)	Passive shoulder vests; powered hip-knee orthoses; clutch-switched knee braces
Structure	**Rigid** (links and joints aligned with the body's joints); **Soft exosuit** (textiles and cables that apply tension across joints with no rigid frame)	Rigid gait exoskeletons; cable-driven soft ankle and hip suits
Application	Medical rehabilitation (clinic); personal mobility (home and community); industrial / occupational; military and first responder; consumer and sport	—

The actuation choice dominates cost, mass and regulation. Passive industrial devices weigh about 2–5 kg, need no battery and sell for a few thousand US dollars. Powered medical exoskeletons weigh 20–30 kg with battery, are regulated as medical devices and cost tens of thousands of dollars or more.

3. Historical development

The idea is 136 years old, but working products arrived only after 2010, when lithium-ion batteries, compact brushless motors and cheap embedded computing matured together. The history falls into four eras.

3.1 Mechanical concepts (1890–1960)

In 1890 the Russian engineer Nicholas Yagn received US patent US420179 for an *Apparatus for Facilitating Walking, Running and Jumping*. It used long leaf springs running parallel to the legs to store energy at foot strike and release it at push-off; it was entirely passive. [1] Later passive designs in the early twentieth century followed the same principle with smaller, more limited mechanisms.

3.2 Powered amplifiers and the first failures (1960–1990)

In the 1960s the US Army and Navy funded General Electric, led by Ralph Mosher, to build **Hardiman I**, a full-body hydraulic "man amplifier" meant to let one operator lift about 680 kg (1,500 lb). The machine itself weighed about 680 kg and measured about 1.8 × 1.0 × 0.7 m. [2] GE's final report of 30 August 1971 documented the programme. [3] Only a single arm ever worked reliably; whole-body operation produced violent, uncontrolled motion. The lesson was that the hard problem is not force but **human–machine coordination**: without fast, precise control the suit was too bulky and dangerous to use. [4] GE reused the hydraulic master–slave work in its quadruped *Walking Truck*.

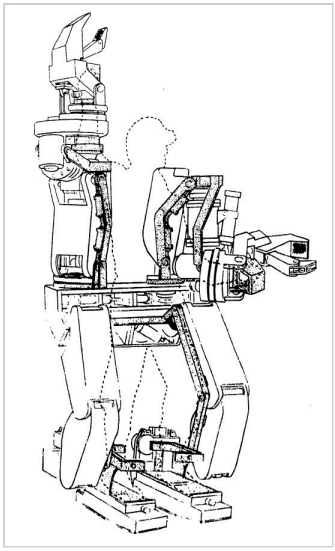


Figure 1. Drawing of General Electric's Hardiman I, the full-body hydraulic "man amplifier" built for the US Army and Navy, from the programme's 1971 final report.

Bruce R. Fick and John B. Makinson, General Electric · Public domain · Wikimedia Commons

In parallel, Mihajlo Vukobratović's group at the Mihajlo Pupin Institute in Belgrade built pneumatically powered lower-limb exoskeletons for paraplegic patients in the late 1960s and early 1970s. Their balance work led to the zero-moment-point (ZMP) concept that still underpins biped and exoskeleton gait control.

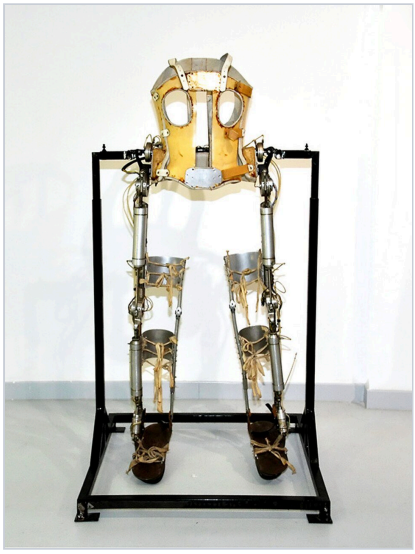


Figure 2. Active exoskeleton for paraplegic patients built at the Mihajlo Pupin Institute, Belgrade, 1972.

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3.3 Defence-funded renaissance (1990–2010)

In 1992 Yoshiyuki Sankai at the University of Tsukuba began the work that became **HAL (Hybrid Assistive Limb)**, among the first exoskeletons with real market potential. [3] HAL reads bioelectric (surface EMG) signals at the skin to detect the wearer's intent before the joint moves.



Figure 3. Cyberdyne's HAL (Hybrid Assistive Limb) on display; it reads surface EMG signals to detect the wearer's intent (photo 2010).

Yuichiro C. Katsumoto · CC BY 2.0 · Wikimedia Commons

In 2000 DARPA launched the **Exoskeletons for Human Performance Augmentation (EHPA)** programme, a US$50 million, five-year effort. [5] From 23 proposals it selected UC Berkeley, Sarcos Research, Oak Ridge National Laboratory and Millennium Jet; Sarcos and Oak Ridge were to develop full-body suits. [5] The target was a suit that let a soldier carry roughly double the 50 kg an unaided soldier is expected to carry, and march faster and longer. [6]

Two results shaped the industry:

- **BLEEX** (Berkeley Lower Extremity Exoskeleton, H. Kazerooni) carried a substantial external load plus its own weight while walking, energetically autonomous, in 2005–2006. [3] BLEEX became the base for **HULC** (Human Universal Load Carrier), licensed to Lockheed Martin, and the spin-out company that became **Ekso Bionics**. [7]
- **Sarcos XOS / XOS 2** (later under Raytheon) demonstrated full-body powered strength augmentation, with the wearer lifting heavy loads repeatedly. It was tethered to an external power supply, which limited field use.



Figure 4. HULC (Human Universal Load Carrier), the load-carriage exoskeleton derived from Berkeley's BLEEX work, shown to US Army officers.

US Army RDECOM · Public domain · Wikimedia Commons



Figure 5. A US Army "Future Soldier" powered-armour concept on public display, 2007: the full-suit vision that later programmes such as TALOS pursued.

Daren Reehl, US Army · Public domain · Wikimedia Commons

3.4 Commercial era (2010–present)

The focus shifted from soldiers to patients and workers, where the value case was clearer and the duty cycle shorter.

- ReWalk (Argo Medical Technologies, Israel) entered clinical trials at MossRehab, Philadelphia, in 2009; it was cleared for clinical use in 2011 and trialled for home use from 2012. [8] It reached the European market in 2012. [7]
- On 26 June 2014 the US FDA granted ReWalk Personal a de novo classification, the first powered exoskeleton cleared for home and community use. [8]
- Ekso Bionics, Parker Hannifin (Indego) and Cyberdyne followed with FDA clearances (Section 5).
- From about 2015 automotive and aerospace plants (Ford, Toyota, BMW, Airbus, Boeing) began deploying passive shoulder and back exoskeletons at scale.
- The US SOCOM TALOS programme (2013–2019) tried to build a powered armoured suit for special forces and ended without fielding one, a second Hardiman-style lesson about power and integration. Military interest moved to lighter, task-specific devices.
- From about 2019 soft exosuits (Harvard Wyss Institute, ReWalk ReStore) and, from 2023, consumer hiking exoskeletons from Chinese and US firms opened new segments.

Year	Milestone	Significance
1890	Yagn walking-assist patent	First exoskeleton patent; passive springs
1965–1971	GE Hardiman I	First powered full-body suit; control failure
c. 1970	Vukobratović, Pupin Institute	Powered gait assistance for paraplegia; ZMP
1992	HAL research begins, Tsukuba	EMG intent detection
2000	DARPA EHPA launched	US$50 M military push
2004–2006	BLEEX, Sarcos XOS	Load-carrying and full-body demos
2009	HULC licensed to Lockheed Martin	Untethered military load carrier
2012	ReWalk CE marked	First commercial medical walking exoskeleton
2014	ReWalk FDA de novo	First home-use clearance
2015 onward	Industrial passive exos in auto plants	Volume market begins
2013–2019	SOCOM TALOS	Powered combat suit cancelled
2019 onward	Soft exosuits, consumer exos	Lighter, cheaper, wider adoption

4. Enabling technologies

A practical exoskeleton is a balance of six subsystems. Weakness in any one makes the device heavier, slower or less trusted than the user's own body.

Subsystem	Current practice	Key engineering trade-off
Actuation	Brushless DC motors with harmonic or planetary reduction (medical); series-elastic actuators for compliance; Bowden-cable drives (soft suits); hydraulics for very high force (heavy-lift, legacy military); springs, gas struts and elastic bands (passive)	Torque density vs back-drivability. High gear ratios give torque but resist natural motion; quasi-direct-drive motors are transparent but heavy
Sensing	Joint encoders, IMUs on thigh/shank/trunk, foot-pressure insoles, load cells; surface EMG (HAL); research use of EEG / brain–computer interfaces	Robust intent detection with few sensors and no skin preparation

Subsystem	Current practice	Key engineering trade-off
Control	Finite-state gait machines triggered by trunk tilt (ReWalk); impedance / admittance control; EMG-proportional assist; human-in-the-loop optimisation that tunes assist profiles to minimise metabolic cost; machine-learning gait-phase estimators	Transparency (no resistance when not needed) vs stability and safety
Power	Li-ion packs of roughly 0.1–0.5 kWh; swappable batteries; passive devices need none	Run time vs added mass. Power density remains the main barrier for military full-body suits
Structure and materials	Aluminium and titanium frames, carbon-fibre composites, technical textiles; adjustable link lengths	Stiffness for load transfer vs mass and comfort
Human interface	Straps, cuffs, padded harnesses; self-aligning joint mechanisms to accommodate joint-axis misalignment	Load transfer to the body without pressure sores, chafing or joint misalignment forces

Two developments since 2015 changed what was possible. Soft exosuits showed that useful assistance (reductions of the order of 10–20 % in metabolic cost of walking in laboratory studies) does not need a rigid frame. Human-in-the-loop optimisation showed that the right assist *profile* matters more than peak force, and that it differs between individuals.

5. Path to commercialisation

The products that reached market did so by narrowing the problem: one user group, one indication, one task, a defined payer. Four mechanisms made that possible.

5.1 Regulatory clearance (medical)

In the US, powered lower-limb exoskeletons are Class II devices under FDA product code PHL. ReWalk's 2014 de novo grant (K131798) created the category; later devices cleared through the 510(k) route with ReWalk as predicate. [9]

Year	Device (maker)	FDA route	Indication / setting
2014	ReWalk Personal (ReWalk Robotics, now Lifeward)	De novo, K131798	SCI; home, community and clinic
2016	Ekso GT (Ekso Bionics)	510(k)	SCI and stroke; rehabilitation clinic
2016	Indego (Parker Hannifin; now Ekso Bionics)	510(k)	SCI T7–L5, with trained companion; clinic and personal
2017	HAL for Medical Use, Lower Limb (Cyberdyne)	510(k), K171909	Rehabilitation
2020	Keeogo (B-Temia)	510(k), K201539	Mobility assistance

Indego is cleared for people with SCI at T7 to L5 to walk with supervision of a trained companion, and also for rehabilitation institutions. [9] In Europe, CE marking under the Medical Device Regulation (EU 2017/745) is the equivalent gate; ReWalk was sold in Europe two years before its US home-use clearance. [7]

5.2 Reimbursement

Clearance alone did not create a market; payment did. After CMS placed personal exoskeletons in the Medicare **brace** benefit category, it set a lump-sum purchase fee in April 2024, with a ceiling-to-floor range of US$109,238 to US$81,929 and a rate of US$91,032 in each state. [10] The rate applies to claims from 1 April 2024 and covered the ReWalk Personal and the Ekso Indego Personal. [11] Before this,

payment was case by case; only 35 ReWalk Personal systems had been delivered to Medicare beneficiaries by February 2024. [12] Germany's statutory insurers and the US Veterans Affairs programme had earlier been the main payers.

5.3 Standards (industrial and general)

Industrial exoskeletons are not medical devices and, in most jurisdictions, are not formally personal protective equipment either, which left buyers without a benchmark. Two standards bodies filled the gap:

- **ISO 13482:2014** (*Robots and robotic devices — Safety requirements for personal care robots*) covers "physical assistant robots", including restraint-type wearable devices.
- **ASTM F48** (*Exoskeletons and Exosuits*), established in autumn 2017, develops standards for wearable devices that assist or enhance motion, posture or physical activity. [13] Its scope spans industrial, emergency response, medical, military and consumer uses, passive and active. [14] Its first two published standards were F3323 (terminology) and F3358 (labelling and information). [15]

The committee's view is that standards and certification would speed workplace adoption, because the lack of them has been a barrier for industry buyers. [16]

5.4 Business models

- **Capital sale to clinics** with training and service contracts (Ekso, Hocoma, Cyberdyne). Clinics justify the price by therapy throughput.
- **Rental and leasing**, including Cyberdyne's HAL rental model in Japan and exoskeleton-as-a-service offers for industrial pilots.
- **Volume sale of passive industrial units** at a few thousand dollars each, sold on reduced injury claims and fatigue rather than productivity.
- **Defence R&D contracts and trials**, usually with no production order; few military programmes have yet moved to fielding.
- **Direct-to-consumer** retail of low-cost hip-assist devices, the newest model (Section 6.4).

6. Major players today (October 2026)

The industry is fragmented: no company dominates all segments, and many firms are small, venture-funded and changing ownership. Revenue is concentrated in medical rehabilitation; unit volume is shifting fast to consumer devices, led by Chinese makers. Company status below was checked against news to October 2026; product lines change often, so readers should confirm current models on the linked sites.

6.1 Medical rehabilitation and personal mobility

Company (country)	Key products	Type	Notes
Lifeward (Israel/USA), formerly ReWalk Robotics	ReWalk Personal, ReStore soft exo-suit	Powered hip-knee; soft ankle exosuit	First FDA home-use exoskeleton (2014); Medicare-reimbursed. Portfolio also includes AlterG and MyoCycle [11]
Wandercraft (France)	Atalante X (clinic), Eve (personal)	Powered, self-balancing, hands-free	Atalante X cleared for stroke (Dec 2022) and SCI T5–L5 [17]. Eve cleared by FDA for personal use, announced 11 Aug 2026 [18]

Company (country)	Key products	Type	Notes
Ekso Bionics (USA)	EksoNR (clinic), Ekso Indego Personal, EVO (industrial)	Powered lower-limb; passive shoulder	Pioneer from Berkeley BLEEX. Parent company is now ChronoScale, which in 2026 classified the exoskeleton business as held for sale [19]
Cyberdyne (Japan)	HAL Lower Limb, HAL Lumbar	Powered, EMG-triggered	FDA 510(k) 2017 [9]; covered by Japanese national insurance for some neuromuscular diseases
DIH / Hocoma (Switzerland)	Lokomat, Armeo	Treadmill-based robotic gait and arm trainers	Stationary rehabilitation robots; long-time clinical leader [20]
Fourier (China)	ExoMotus series	Powered lower-limb rehab	Large Chinese rehab-robot maker
Myomo (USA)	MyoPro	Powered upper-limb orthosis, EMG	Arm and hand function after stroke or brachial plexus injury
B-Temia (Canada)	Keeogo	Powered knee assist	FDA 510(k) 2020 [9]
Hyundai Motor Group Robotics Lab (S. Korea)	X-ble MEX	Powered lower-limb rehab	Launched June 2023 [20]

Wandercraft's Eve is the first self-balancing exoskeleton for personal use; users do not need crutches. Its clearance rested on a three-site trial, and Medicare already pays for personal exoskeletons under billing code K1007. [18]



Figure 6. A user walking in an EksoNR rehabilitation exoskeleton with a therapist.

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Figure 7. US Marine Capt. Derek Herrera (centre), paralysed in combat in 2012, standing in an exoskeleton at his retirement ceremony, Camp Pendleton, 2014.

Sgt. Scott Achtemeier, US Marine Corps · Public domain · Wikimedia Commons

6.2 Industrial and occupational

Company (country)	Key products	Type	Notes
Ottobock Bionic Exoskeletons / suitX (Germany/USA)	Paexo Shoulder, Paexo Back, suitX range	Passive shoulder, back, wrist, neck	Paexo launched after a 30-worker Volkswagen Bratislava trial; priced at about €5,000 at launch [21]. Ottobock bought suitX in 2021 [22]
Hilti (Liechtenstein)	EXO-O1, EXO-S	Passive overhead-work shoulder	Developed with Ottobock for construction [23]
Comau (Italy)	MATE-XT	Passive shoulder	Automotive-plant origin [20]
Levitate Technologies (USA)	AIRFRAME	Passive shoulder	Trialled at BMW Spartanburg [21]; tunnel construction at Granite Construction [24]

Company (country)	Key products	Type	Notes
Laevo (Netherlands)	Laevo Flex	Passive back	Logistics and warehousing
Hyundai Motor Group (S. Korea)	X-ble Shoulder	Passive shoulder	Used in Hyundai and Kia plants
Verve Motion (USA)	SafeLift	Powered soft back exosuit	Harvard soft-exosuit lineage; venture-backed [24]
HeroWear (USA)	Apex 2	Passive soft back	Commercial version of the Army/Vanderbilt SABER exosuit [25]
German Bionic (Germany)	Apogee	Powered, AI-connected back assist	Filed for bankruptcy in November 2025 [26]

Passive shoulder and back devices are the volume product. Powered industrial suits have struggled commercially: German Bionic's insolvency and Sarcos's exit from its Guardian XO heavy-lift suit (the company reorganised as Palladyne AI in 2024) show how hard it is to make a powered industrial suit pay back on the shop floor.



Figure 8. Comau's passive MATE-XT shoulder exoskeleton in a logistics task.

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Figure 9. A Puget Sound Naval Shipyard worker grinding metal with a backpack-carried tool-support exoskeleton, 2011.

Alan Baribeau, US Navy · Public domain · Wikimedia Commons

6.3 Military and defence

Organisation (country)	Programme / product	Type	Status
Lockheed Martin (USA)	HULC (2009); ONYX knee exoskeleton with Dephy; FORTIS tool-arm exo	Powered knee; passive tool support	ONYX developed under a two-year sole-source agreement with US Army Natick [27]; FORTIS used B-Temia technology [28]
US Army DEVCOM / Futures Command	SABER (Pathfinder, with Vanderbilt); infantry exoskeleton requirement	Passive soft back; powered lower-limb	SABER reduced lift back strain by up to 40 % but was not adopted [25]; Army formally pursued an infantry exoskeleton from 2020 [29]
Ottobock suitX (USA)	Army evaluation contract	Powered	Commercial units tested at Fort Sill, late 2024 [25]
Russia	K-2 passive lower-body; Ratnik-3 concept	Passive; powered concept	K-2 shown 2017; Ratnik-3 shown 2018 [29]
China (PLA)	Logistics and high-altitude load-carriage suits	Passive and powered	PLA observed using suits for supply, logistics and patrol [30]

Organisation (country)	Programme / product	Type	Status
Tata Advanced Systems + DRDO (India)	Full-body gravity-redistribution exoskeleton for infantry	Passive	Unveiled 2025 [31]; reported to transfer up to 75 % of carried weight to the ground [32]
Svaya Robotics + DRDO R&DE Pune and DEBEL Bengaluru (India)	Active load-carriage exoskeleton	Powered	Technology demonstrator sized to Indian soldier anthropometry [33]

Military interest has moved away from full "Iron Man" suits toward three narrower jobs: load carriage on the march (knee and hip assist), logistics lifting (back exosuits) and tool or weapon support (passive arm exos). DRDO is developing both passive and powered exoskeletons for logistics tasks, after capturing and analysing soldier biomechanics for those tasks. [30]

6.4 Consumer and sport

Company (country)	Key products	Price band	Notes
Hypershell (China/HK)	X Go, X Pro, X Carbon, X Ultra S; Halo	Entry to premium; Halo Origin €2,299	Top-selling consumer brand, sold in over 70 countries; US$50 M Series B+ in May 2026 [34]. Halo (hip + knee, 2.6 kg) debuted at IFA 2026 [35]
Dnsys (China)	X1 Lite, X1 Carbon, X1 Carbon Pro	Entry to mid	Powered hip [36]
RoboCT (China)	GoGo series	Entry	Moved from hospital rehab to consumer; targets 60,000–100,000 units in 2026 [37]
Samsung (S. Korea)	Bot Fit (formerly GEMS Hip)	Mid	Hip-assist for walking exercise
Skip with Arc'teryx (USA/Canada)	MO/GO	Premium	Powered knee for hiking, spun out of Google X
Roam Robotics (USA)	Pneumatic knee exos	Premium	Ski and knee-assist; also military and medical variants

China shipped about 26,000 exoskeletons in 2025 worth over RMB 1.6 billion. Consumer devices were about 73 % of units but only about 7 % of revenue; medical rehabilitation was about 13 % of units and 88 % of revenue. [38]

7. Market size and trends

The global exoskeleton market was worth roughly US$0.56–0.60 billion in 2025, and analysts expect it to reach about US$1.8–2.3 billion within eight to ten years. Estimates differ widely because firms define the market differently (some include stationary rehab robots or count consumer devices, some do not).

Source (published)	Scope	Base value	Forecast	CAGR
Towards Healthcare (Mar 2026)	All exoskeletons	US$595 M (2025)	US$2,309 M (2035)	14.5 %
Research and Markets (Feb 2026)	All exoskeletons	US$590 M (2025)	US$1,790 M (2033)	14.5 %
MarketsandMarkets via GlobeNewswire (Jun 2025)	All exoskeletons	US$560 M (2025)	US$2,030 M (2030)	29.4 %
Straits Research (May 2026)	Military only	US$200.6 M (2025)	US$811.5 M (2034)	16.8 %

Five trends stand out:

- **Volume is moving to consumers.** Consumer hip-assist devices now ship in the tens of thousands per year and are sold in Prime Day-style retail promotions, a sign the category is maturing. [39]

- **China is the volume manufacturer.** Low-cost supply chains for motors, batteries and drives have let Chinese brands lead consumer sales and start price wars. [38]
- **Medical revenue depends on reimbursement.** The 2024 Medicare rate and the 2026 Eve clearance make home-use personal exoskeletons a reimbursable product in the US for the first time at scale.
- **Consolidation and failures are common.** Ekso's parent has put the exoskeleton business up for sale, and German Bionic went bankrupt; the field still has more companies than profitable niches.
- **AI-based control** is replacing rule-based gait state machines; Hypershell, for example, markets end-to-end learned motion control with independently verified gait-synchronisation figures. [40]

8. Challenges and limitations

The Hardiman lesson still holds: the limiting factors are power, control and the human interface, not raw force.

1. **Power density.** Batteries give hours of assist for a hip device but not for a full-body powered suit doing heavy work. Every kilogram of battery is carried by the user or must be supported by the frame. This is the main reason powered military suits (Hardiman, XOS, TALOS) never fielded.
2. **Human–machine interaction and intent.** The suit must know what the user wants to do before or as they do it. Misreading intent on stairs, slopes or a stumble can be dangerous; over-assistance can cause users to lose muscle strength.
3. **Fit and joint alignment.** Human joints are not simple hinges; the knee's axis moves as it bends. Misalignment causes chafing, pressure points and parasitic forces. Fitting one design to the full range of body sizes, including women and smaller-statured populations, remains difficult.
4. **Evidence of benefit.** Laboratory metabolic savings do not always carry over to real tasks. For industrial devices, long-term injury-reduction data are still limited, and some studies show load shifted to other body regions rather than removed.
5. **Safety and standards.** Industrial devices fall between medical-device and PPE rules. ASTM F48 and ISO 13482 help, but buyer-specific certification and testing frameworks are still forming. [16] Consumer devices raise new questions about falls, misuse and inflated specifications.
6. **Cost and business case.** Medical exoskeletons cost around US$80,000–110,000 (the 2024 Medicare range), [10] limiting access outside reimbursed markets such as India. Industrial buyers want payback within one to two years.
7. **User acceptance.** Workers reject devices that are hot, slow to don, catch on structures or make them look different. Soldiers reject anything that adds noise, heat signature or a new failure point.
8. **Supply chain and data security.** Motors depend on rare-earth magnets, an identified risk for military exoskeletons. [39] Connected devices also collect biometric and motion data that need protection; ASTM F48 has sought cybersecurity expertise for this reason. [15]

9. Future outlook

Over the next decade exoskeletons are likely to become lighter, cheaper and more specialised rather than more powerful. The winners will be devices that do one job well, are donned in under a minute and have a clear payer. The outlook below is the authors' assessment, drawing on the trends in Sections 6–8.

Application	Near term (2026–2030)	Longer term (2030–2036)	What must happen
Medical rehabilitation	Wider clinic use for stroke, SCI, MS; AI-tuned gait therapy; lower-cost Chinese and Korean devices enter Asian markets	Exoskeleton gait training becomes routine; combined with functional electrical stimulation and brain–computer interfaces	Stronger outcome evidence; reimbursement outside the US, Germany and Japan
Personal mobility (SCI, elderly)	Self-balancing home devices (Eve) scale through Medicare and distributors; hip and knee assist for frail elderly	Lightweight soft suits for ageing populations in Japan, Korea, China, Europe; partial replacement of wheelchairs for some users	Service networks, training, falls-safety standards, insurance coverage
Industrial	Passive shoulder and back devices become standard PPE-like equipment in auto, aerospace, construction, logistics; powered soft back suits for warehouses	Connected exos with ergonomic data feeding safety programmes; task-specific modular kits	Long-term injury data; ASTM / ISO certification accepted by safety regulators and insurers
Military and first responder	Knee and hip load-carriage aids, logistics lifting suits, passive tool-support arms; trials in high-altitude and mountain operations	Integrated soldier systems sharing power with radios and sensors; hybrid devices that are passive by default and powered on demand	Silent, rugged, low-heat designs; better batteries; doctrine and training that use the devices
Consumer and sport	Hip-assist for hiking, commuting and elderly walking sells in the hundreds of thousands of units; retail channels expand	Exos treated like e-bikes: mass-market, regulated as mobility products; integration with fitness apps	Independent safety and performance testing; truthful specifications



Figure 10. Honda's Walking Assist Device with bodyweight support, shown at the 2013 Tokyo Motor Show: an early step toward everyday mobility aids for older people.

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9.1 Technology directions

- **Soft and hybrid architectures** that weigh 1–3 kg and switch between passive and powered modes.
- **Learned control** (on-device machine learning) that adapts the assist profile to each user and terrain within minutes.
- **Better actuators**: quasi-direct-drive motors, variable-stiffness and clutch-spring elements, and artificial-muscle materials.
- **Neural interfaces**, from improved surface EMG to non-invasive and implanted brain–computer interfaces for people with complete paralysis.
- **Digital twins and biomechanical simulation** to design and certify devices before human trials.

9.2 Relevance to India

India has a large rehabilitation need and a defence requirement for load carriage in high-altitude terrain, but imported medical exoskeletons at US$80,000 or more are out of reach for most users. Indian industry and DRDO laboratories have started building local capability: the TASL–DRDO passive infantry exoskeleton, the Svaya–DRDO active demonstrator, and work by start-ups such as Newndra. [31] The Bureau of Indian Standards held a workshop on exoskeleton safety and use, [31] a first step toward a national framework.



Figure 11. Robotic gait-training demonstration at a neuro-rehabilitation centre in Guwahati, Assam, India.

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The main opportunities are:

- Low-cost passive exoskeletons for soldiers, porters and logistics troops in Ladakh, Sikkim and Arunachal sectors.
- Affordable rehabilitation exoskeletons for district hospitals, built on Indian supply chains.
- Industrial exos for automotive, aerospace MRO and shipbuilding.
- Reliability and endurance testing of drive trains, bearings and gearboxes for exoskeleton actuators.

10. Conclusions

- Exoskeletons took 124 years to go from Yagn's 1890 spring patent to the first home-use clearance in 2014. The delay was caused by control, power and interface problems, not by a lack of mechanical strength.
- Commercial success came from narrowing the problem: medical devices for defined patient groups with a payer, and passive industrial devices for specific tasks.
- In 2026 the market is about US$0.6 billion, led in revenue by medical rehabilitation and in units by Chinese consumer hip-assist devices. Many companies are small, and failures and ownership changes are frequent.
- Military use is converging on light, task-specific load-carriage and logistics aids rather than powered armour.
- The next decade will favour soft, hybrid, AI-controlled devices with credible standards and evidence. India has a clear opportunity in low-cost defence and rehabilitation devices built on domestic supply chains.

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Figure credits

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